

# RW-14 Internal Measurement Rules and Tolerances

Project: RW-14 Oak Lane retaining wall and site drainage improvement

This reference defines calculation methods, tolerance limits, and status handling for the closeout quantity verification package.

| Rule ID | Subject                   | Measurement rule                                                                                                                                       | Units / limit           |
|---------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| MR-001  | Concrete volume           | Calculate retaining wall concrete volume only when length, exposed wall height, and wall thickness are explicitly stated: length x height x thickness. | m3; round to 3 decimals |
| MR-002  | Reinforcement weight      | Calculate reinforcement weight only when bar length and unit mass are explicitly stated: bar_length_m x unit_mass_kg_per_m.                            | kg; round to 2 decimals |
| MR-003  | Drainage aggregate volume | Calculate drainage aggregate volume only when trench length, aggregate width, and aggregate depth are explicitly stated: length x width x depth.       | m3; round to 3 decimals |
| MR-004  | Geotextile area           | Calculate geotextile area only when roll or installed length and roll width are explicitly stated: length x roll_width.                                | m2; round to 2 decimals |
| MR-005  | Excavation volume         | Calculate excavation volume only when length, excavation width, and average depth are explicitly stated: length x width x average_depth.               | m3; round to 3 decimals |
| MR-006  | Elevation variance        | Calculate variance as as-built elevation minus design elevation for the same point ID or explicitly matching wall segment/station.                     | m; round to 0.001       |

| Tolerance ID | Measurement                                  | Threshold                                                       | Status handling                                       |
|--------------|----------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------|
| TL-001       | Top of retaining wall elevation              | absolute variance <= 0.025 m                                    | SOURCE_OK if within threshold; CONFLICT if exceeded   |
| TL-002       | Drain invert elevation                       | absolute variance <= 0.020 m                                    | SOURCE_OK if within threshold; CONFLICT if exceeded   |
| TL-003       | Concrete ticket total versus measured volume | absolute variance <= 0.150 m3 for each wall segment             | SOURCE_OK if within threshold; CONFLICT if exceeded   |
| TL-004       | Concrete 28-day compressive strength         | minimum 35.0 MPa                                                | SOURCE_OK at or above minimum; CONFLICT below minimum |
| TL-005       | Rebar mill certificate reference             | certificate reference must be present for a recorded weight row | MISSING when certificate or report ID is blank        |
| TL-006       | Closeout reference                           | closeout reference must be populated for the final package row  | MISSING when blank                                    |

| Status    | Definition                                                                                                                                            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOURCE_OK | Use when a value is directly supported by a source record and no explicit conflicting record is found.                                                |
| CONFLICT  | Use when another supplied source gives a different value for the same wall segment, station, pay item, survey point, measurement, or tolerance check. |
| MISSING   | Use when a required source value, reference, certificate, or closeout field is blank or absent.                                                       |
| DUPLICATE | Use when the same source file repeats the same station, item, value, and unit.                                                                        |
| TEXT_ONLY | Use for descriptive inspection or handover notes with no numeric quantity.                                                                            |